Homework #8

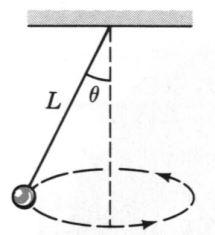
1) The electron in a hydrogen atom orbits a stationary proton at a distance of 5.30×10^{-11} m at a speed of 2.20×10^{6} m/s. Find: (a) the period; (b) the force on the electron.

2) Suppose the coefficient of static friction between the road and the tires on a car is 0.60. What speed will put the car on the verge of sliding as it rounds a level curve of 30.5 m radius?

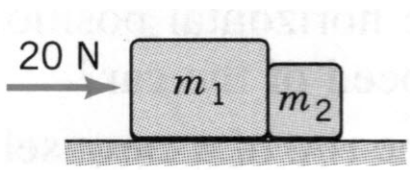
3) A car rounds a circular curve of radius 80.0 m banked at 15° to the horizontal at 30.0 m/s. What is the coefficient of friction required for the car not to skid? Some hints. first, set the x-axis as horizontal, and the y-axis as vertical (this is different than how we normally handle inclines). Second, there will be 3 forces: the normal force (with x- and y- components), the force of gravity (y-component only), and friction pointing down the slope (also with x- and y-components).

4) In a conical pendulum, shown in the figure below, a bob moves in a horizontal circle. Show that the period is

$$T = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$



5) Two blocks, $m_1 = 5.00$ kg and $m_2 = 3.00$ kg, are in contact with each other and are on a horizontal surface for which $\mu_k = 0.200$. A 20.0 N horizontal force acts on the 5.00 kg block as shown in the figure. (a) What is the acceleration of the blocks. (b) What is the force exerted by on m_1 by m_2?



6) A 2.00-kg block is pulled 3.00 m along a frictionless horizontal plane by a 10.0-N force that acts at 37º above the horizontal. What is the work done by the force on the block?

7) A 200-g ball thrown vertically up with an initial speed of 20.0 m/s reaches a maximum height of 18.0 m. Find: (a) the change in its kinetic energy; (b) the work done by gravity. (c) Are the two quantities just calculated equal? Explain why or why not.

8) A 12.0 N force with a fixed orientation does work on a particle as the particle moves through the three-dimensional displacement $\vec{d} = (2.00\,\hat{\imath} - 4.00\,\hat{\jmath} + 3.00\,\hat{k})$ m. What is the angle between the force and the displacement if the change in the particle's kinetic energy is (a) +30.0 J and (b) −30.0 J?